

Operators:

```
int a = 10;
int b = 20;
int sum = a + b;
```

Annotations:
 - `int`: keyword
 - `a`, `b`: identifier/variable
 - `+`: operator
 - `;`: Terminating Semicolon
 - `a`, `b`: operand

Operator: It is a symbol which performs an operation between two or more operands to give a result.

1. Arithmetic Operators: $+, -, /, *, \%$
 C, C++, Java, JS
 Python: $**$, $//$

2. Assignment Operators:
 Augmented: $=, +=, -=, /=, *=, \%=$
 Short-hand: $a = 10$
 $[a = a + 5]$
 $[a += 5]$
 Same

3. Quotient $\rightarrow$ $10 / 3 = 3$
Remainder $\rightarrow$ $10 \% 3 = 1$
 Python: $10 // 3 = 3$
 $10 \% 3 = 1$
 $10 / 3 = 3.33$

* When the smaller number is not divisible by the bigger number, the smaller number is the remainder.

$5 / 6 = 5$
 $4 / 5 = 4$

* Comparison/Relational Operators: $>, <, <=, >=, ==, !=$
 Boolean
 True or False

* Logical Operators: $\&, ||, !$
 AND OR NOT
 Coding Round
 Timing

5 - 0101
 7 - 0111
 8 - 1000

Fast
 X

OFF - 0 ON - 1 $\rightarrow$ 0 & 1 = 2 digits
 2048: 1024 512 256 128 64 32 16 8 4 2 1
 92
 2 | 92 = 46
 2 | 46 = 23
 2 | 23 = 11
 2 | 11 = 5
 2 | 5 = 2
 2 | 2 = 1
 1 - 0
 Mod Signif Bit
 MSB
 Least Significant Bit
 LSB
 1011100
 $64 + 16 + 8 + 4 = 92$
 $80 + 12 = 92$
 0 0 0 0 0
 0 0 0 1 1
 0 0 1 0 2
 0 0 1 1 3
 0 1 0 0 4
 0 1 0 1 5
 1 1 1 1 15

Bit Masking / Bit Manipulation:

AND
 OR
 NOT
 $(2 > 1) || (3 > 4)$
 $(2 > 3) \&\& (3 > 2) = T \&\& F = F$
 $F \&\& T = 0$
 $0 \&\& 1 = 0 \rightarrow F$

Unary Operators:

Increment/Decrement
 Prefix/Postfix
 $++, --$
 C, C++, Java, JS
 Python
 $a++$
 $a = a + 1$
 $a += 1$

* Ternary Operator: $T \rightarrow F$
 $\rightarrow$ Syntax $\rightarrow$ (Condition) ? trueValue : falseValue
 * Shorthand if-else statement
 ECE
 EEE
 M
 C
 Syntax
 CS
 LS
 Syntax

Bitwise Operators:

① AND $\rightarrow \&$
 ② OR $\rightarrow |$
 ③ XOR $\rightarrow ^$
 ④ Right Shift $\rightarrow \gg$
 ⑤ Left Shift $\rightarrow \ll$
 ⑥ NOT / Complement $\rightarrow \sim$
 Negation
 integers $\rightarrow$ Bit Masking
 int & int
 Compiler
 bin AND bin
 int
 Result

5 $\rightarrow$ 0101
 9 $\rightarrow$ 1001
 $5 \& 9 \rightarrow 0001 \Rightarrow 1$
 XOR
 2 $\rightarrow$ 0010
 11 $\rightarrow$ 1011
 $2 \wedge 11 = 9$
 3 0011
 6 0110
 $3 \vee 6 = 7$
 15 $\rightarrow$ 1111
 12 $\rightarrow$ 1100
 $15 \oplus 12 = 3$

Bitwise Shift Operators: (Masking)

* Value $a = 10$
 * Step/Unit = 2 bits
 RS $a \gg 2$
 LS $a \ll 2$
 $8 \gg 3 = 1$
 $8 \ll 3 = 64$
 L $\gg$
 L $\rightarrow$ R
 1000
 00001010
 Added
 $10 \gg 2 = 2$
 32 16 8 4 2 1
 00010110
 Ignored
 $10 \ll 2 = 40$
 Ignored
 00001010
 00000001
 00000000
 00000000